F3Net: Adaptive Frequency Feature Filtering Network for Multimodal Remote Sensing Image Registration

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***Abstract*— Multimodal remote sensing image registration is crucial for multimodal information fusion and applications. The significant nonlinear appearance difference between multimodal images caused by the various imaging mechanisms dramatically increases the challenge of image registration. This article proposes an adaptive frequency feature filtering network (F3Net) for cross-modal remote sensing image registration. On the one hand, F3Net explicitly explores the useful frequency components across modal images based on multilevel deep features. On the other hand, F3Net can take advantage of the nonlocal receptive fields by frequency modulation for feature learning and boosting image registration performances. F3Net inserts frequency feature filtering (F3) modules in multilevel deep features. Specifically, F3Net first performs the fast Fourier transform (FFT) for deep features. Then, F3Net designs a frequency attention (FA) module to adaptive enhance the shared and discriminative frequency features between multimodal images while suppressing the frequency components that hinder the cross-modal image registration. In addition, F3Net adopts multiscale frequency filtering fusion to facilitate discriminative feature learning, including global frequency feature filtering (GF3) based on the global image spectrum and local frequency feature filtering (LF3) based on the spectrum of stacked image regions. Experimental results on many remote sensing images have demonstrated the efficiency of the F3Net on multimodal image registration.**

***Index Terms*— Frequency features, frequency filtering, image registration, multimodal image.**

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I. Introduction

REMOTE sensing image registration is crucial for multimodal image fusion, multitemporal image analysis, and multiview image application. Image registration mainly aligns the images in the same geometric space [1], [2], [3]. Cross-modal image registration aims to register the images acquired by different sensors, such as optical and near-infrared (NIR) images or optical and synthetic aperture radar (SAR) images. Multimodal images contain complementary information, which can provide rich colors and textures for robust image perception under complex environments, such as weak illumination or night, cloud occlusion, and foggy interference. Thus, cross-modal image registration is essential for multimodal image information fusion applications. However, due to the multimodal images having different imaging mechanisms, there are significant appearance differences between multimodal images [4]. It is challenging to achieve accurate cross-modal image registration performances.

Traditional image registration methods try to align images by comparing the image similarity based on the raw image intensity or building local correspondences based on the low-level handcrafted features, i.e., gradient features [5], shapes [6], and frequency information [7]. The former improves the image registration performance from the perspective of image similarity measurement. For example, the widely used image similarity metrics are mutual information (MI) [8], spatial and MI (SMI) [9], normalized cross correlation (NCC) [10], and the sum of square differences (SSD). The latter improves image registration results from the perspective of robust feature extraction. In addition, several works combine multiple features or multiscale features to enhance the robustness of features [11]. The remote sensing data characteristics can also be used to guide the feature extraction process and boost remote sensing image registration performance [12], [13]. To further improve the registration accuracy, multistage coarse-to-fine registration methods progressively achieve more

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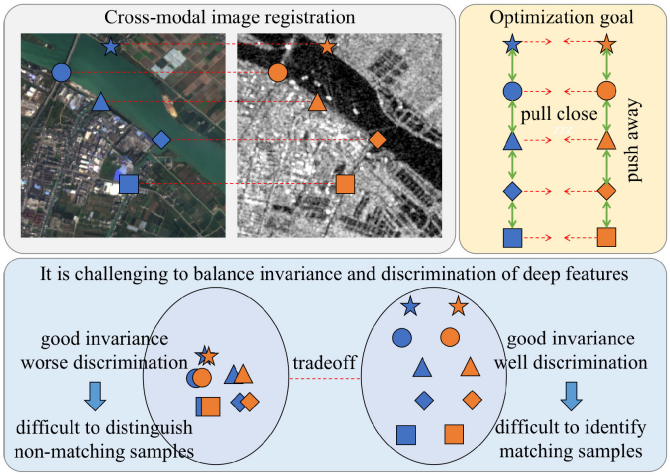


Fig. 1. Deep learning methods are challenging to balance the invariance and discrimination of features for image matching.

accurate registration results through the iterative mode [14], [15], [16]. However, traditional methods are sensitive to image changes. Their registration performance on cross-modal images should be further improved.

Deep learning methods mainly extract high-level features from local image patches or whole images for image registration [17], [18], [19]. The former extracts the local descriptor surrounding the keypoints by the deep convolutional neural network (CNN) and builds the local correspondences between the images for image transformation parameters' estimation and registration. The latter extracts the deep features from the whole image and then finds the matching localization with the maximum similarity using the template matching strategy. The deep feature learning methods expect to identify the matching position and distinguish nonmatching positions. However, the images may contain various image changes, such as illumination changes, geometrical transformations, and image modality changes. Thus, the extracted features should be invariant to these image changes, where the features of matching points are similar. Meanwhile, the features should have strong discrimination to separate the nonmatching points from the matching points. To deal with the image modality differences in cross-modal image registration, the deep network forces the features of the matching samples to be close, which may discard some detailed information and affect the discrimination of the features for distinguishing the nonmatching samples. As shown in Fig. 1, it is difficult to distinguish the nonmatching samples when the deep features have good invariance but worse discrimination. In contrast, it is difficult to identify the matching samples when the deep features have worse invariance but good discrimination. Deep learning methods are challenging in balancing the invariance and discrimination of the features for image matching and registration.

Existing works find that frequency components have different preferences in the CNN learning process [20]. This also means that frequency components have different sharing abilities in cross-modal images. Inspired by this, this article proposes an adaptive frequency feature filtering network (F3Net) for cross-modal remote sensing image registration. F3Net can explicitly explore the useful frequency components of different sharing abilities across modal images. In addition, the frequency modulation will perform the global affections in all input data [21]. Thus, F3Net can combine the advantages of the nonlocal receptive fields through frequency modulation in the feature learning process, boosting image registration performances.

To achieve this, F3Net first performs the fast Fourier transform (FFT) for deep features. Then, F3Net designs a frequency attention (FA) module to adaptive select the frequency feature with high sharing ability between multimodal images and discrimination to improve the cross-modal image registration. Moreover, F3Net adopts multiscale frequency filtering fusion to enhance discriminative texture feature learning. Specifically, F3Net performs global frequency feature filtering (GF3) based on the global image spectrum and local frequency feature filtering (LF3) based on the spectrum of stacked image regions. After that, the spatial features, global frequency-filtered features, and local frequency-filtered features are fused for image registration by template matching.

The main contributions of this article are given as follows.

1) This article proposes an adaptive F3Net for cross-modal remote sensing image registration. We explore the deep sharing features across multimodal images from a frequency perspective.
2) F3Net designs a multiscale feature fusion for more accurate image registration, which combines the spatial features and global and local frequency-filtered features to enhance the discriminative features.
3) Experimental results on many multimodal images have shown the effectiveness of the proposed F3Net on remote sensing image registration.

II. Related Work

A. Remote Sensing Image Registration

1) Traditional Methods for Image Registration: Image registration methods can be categorized into intensity- and feature-based approaches. The intensity-based method computes the matching parameters by searching for the maximal image similarity based on the image intensity information. The core of the intensity-based method is how to measure image similarity. The widely used similarity measures include the SSD, NCC [10], and MI [8], [22]. SSD calculates the intensity difference between two images. NCC normalizes the cross correlation of image intensity. SSD and NCC are sensitive to image changes and nonlinear radiometric differences between images, which are unsuitable for multimodal image registration. MI measures image similarity from the perspective of informational theory by computing the joint and marginal probability distributions. Therefore, MI exhibits stronger robustness to image changes. However, MI does not consider spatial information and is unstable in the template-matching process.

Feature-based methods align images by extracting and matching image salient features. Traditional features are related to handcrafted features, such as gradient features, texture features, and shape features. The widely used handcrafted features are Harris corners [23], scale invariant feature transform (SIFT) [5], and SIFT variants [12], [13], [24].

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SIFT [5] detects keypoints at multiple scales and generates the descriptor vector with rotation-invariant and scale-invariant. Speeded-up robust features (SURF) [24] accelerates the SIFT and enhances the stability of features to image illumination changes. OS-SIFT [12] and SAR-SIFT [13] consider the characteristics of SAR images and modify SIFT for more accurate SAR image registration. These handcrafted features can deal with slight image changes (i.e., scale change, rotation transformation, and illumination change) and perform well in homogenous image registration. However, handcrafted features cannot achieve accurate results due to the nonlinear radiometric differences between multimodal images. To improve the multimodal image registration performance, Ye and Shen [25] propose the histogram of oriented phase congruency (HOPC) descriptor based on the structural properties of images. After that, they adopt the NCC to measure the structural similarity between images based on the extracted HOPC descriptor. Cui and Zhong [26] design a multiscale phase congruency descriptor (MS-PC) to capture the shape and structural features of images, which can decrease the sensitivity of descriptors to radiometric differences. Ye et al. [27] propose channel features of orientated gradients (CFOG). It utilizes the pixelwise feature representation based on the orientated gradients of images for image structural similarity representation and template matching. However, handcrafted features have limited adaptability and generalization and are usually suitable for specific tasks.

2) Deep Learning Methods for Image Registration: Deep learning methods use the deep network to extract high-level features from images, which are robust to image changes and perform significantly better than handcrafted features. The deep learning methods expect the matching samples to have higher similarity than nonmatching samples in the feature space. They can be divided into two classes: deep local feature learning methods and deep global feature learning methods.

Deep local feature learning methods mainly extract the local features from image patches and then build local correspondences across images for matching and registration. HardNet [28] adopts a deep convolutional network to extract local descriptors from image patches. It optimizes the deep network through a triplet loss and the hard sample mining strategy. Deep feature correlation learning network (CNet) [18] inserts an attention module into the feature extraction stage to enhance feature representation. Feature decomposition network (FDNet) [29] decomposes the features of cross-modal images into modality-shared and modality-specific features. The shared features are then used for matching, alleviating the negative impact of significant differences between multimodal images. Xiang et al. [30] also introduce a feature decoupling network that decouples the semantic features of optical and SAR image registration from the noise information, effectively removing speckle noise, preserving more useful information, and improving registration accuracy. Self-distillation feature learning network (SDNet) [19] proposes a self-distillation feature learning network for optical-SAR image registration, which provides rich supervised information based on more fined similarity for boosting deep network training. Ye et al. [31] combine the high-level features learned by the CNN and low-level features extracted by SIFT for image registration. Deep local feature learning approaches are robust to noise and image changes, acquiring significant performances on remote sensing image matching and registration. However, because of the restricted perception of image content within image patches, deep local feature learning methods fail to capture abundant context information. They are susceptible to similar local pattern interference and confusion, resulting in many false matching point pairs. For images with many repetitive local textures or have poor textures, their matching performances need to be further improved.

Deep global feature learning methods extract the global features from the input whole images and find the matching position through template matching, where the matching location has the highest similarity. MCGF [32] proposes multiscale convolutional gradient features (MCGFs) for multimodal image matching by a fast template scheme. MCGF combines the multiorientated gradient features and multiscale deep convolutional features. Global feature-based template matching method (GFTM) [33] utilizes a deep convolutional network to extract global features from multimodal images and performs fast template matching based on the extracted global deep features. Deep global feature learning methods can utilize rich context features for more accurate image matching and registration. However, deep global feature extraction requires a large amount of computation costs and high storage requirements. In practice, it is necessary to consider reducing the channel dimension of the global feature and speeding up the feature extraction process and similarity computation.

3) Discussion: Existing deep feature learning methods aim to eliminate the significant differences between multimodal images and learn modality-invariant features to identify matching samples while capturing discriminative features to separate the nonmatching samples. However, it is challenging to tradeoff between invariance and discrimination in the feature learning process. This article proposes an F3Net for cross-modal remote sensing image registration, which improves the matching performances from the perspective of frequency modulation. Deep wavelet learning network (DW-Net) [34] also improves the remote sensing image matching and registration from the frequency perspective. It combines high-level convolutional features and spectral features for more accurate cross-modal image matching and registration. Unlike DW-Net performing the wavelet transform on the input images, this article conducts the frequency transform on the multilevel convolutional feature. In addition, DW-Net aims to fuse the deep convolutional and spectral features to enhance the registration performances. This article explores the useful frequency components in deep features by frequency filtering. Thus, there are significant differences between the proposed method and existing methods.

Compared with existing deep learning methods, the main advantages of frequency feature filtering for multimodal remote sensing image registration are given as follows. First, F3Net explores the useful frequency components of different sharing abilities across modal images. It enhances feature learning by adaptively selecting the frequency feature with a high sharing ability between multimodal images and discrimination. Second, F3Net can take advantage of the nonlocal receptive fields through frequency modulation, boosting image

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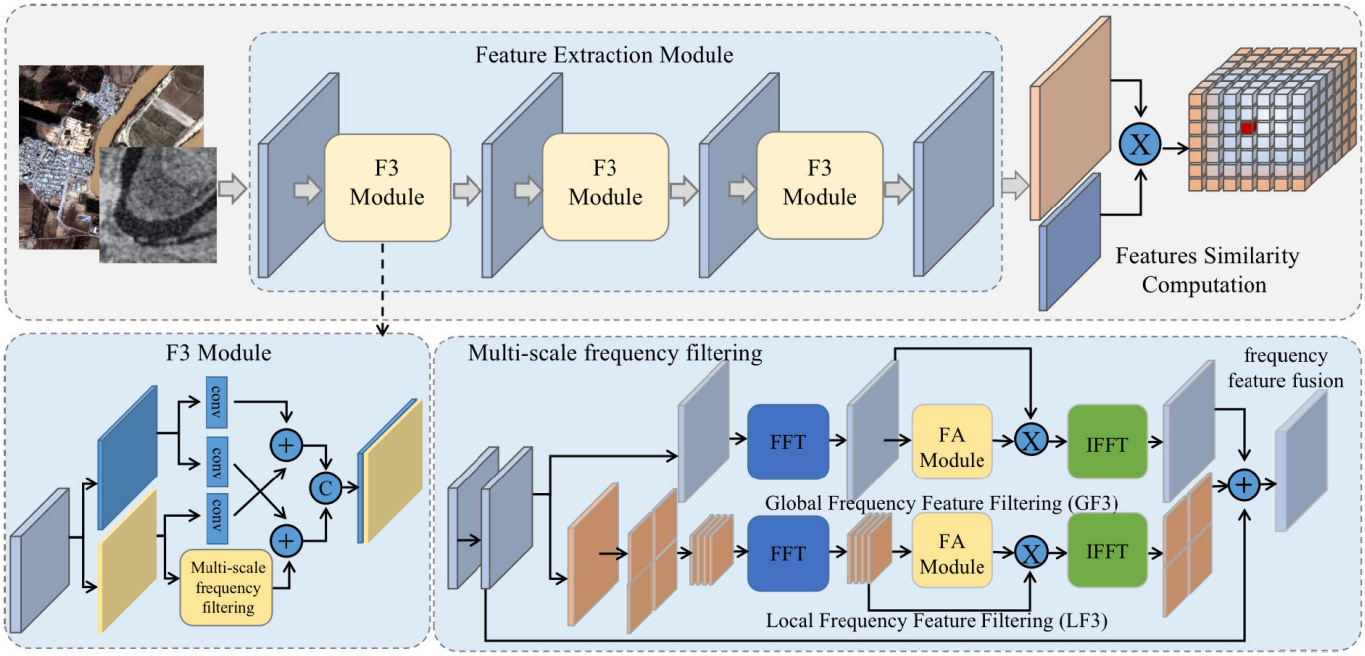


Fig. 2. Framework of the proposed F3Net. It inserts several F3 modules into the multilevel deep features and conducts image template matching via the similarity computation of the frequency-filtered deep features.

registration performances. In addition, F3Net combines the spatial and multiscale frequency features for improving image matching and registration.

B. Frequency Analysis

Frequency information analysis is crucial for image processing. The frequency phase information contains many essential features of a signal [35] and can be used for image reconstruction well. Existing frequency transform approaches for image processing mainly leverage several advantages of frequency information.

First, phase information is more robust than image intensity. Thus, the Fourier transform-based image processing methods demonstrate strong robustness to noise and high invariance to image contrast and brightness changes [36], [37]. Second, image style and content information can be decomposed in the frequency domain, boosting the performance of domain adaptation. Low-frequency components of the image reflect the style information of the image, while the high-frequency spectrum contains image content and semantic information [38]. Therefore, Fourier domain adaptation (FDA) [39] performs image domain alignment by swapping the low-frequency spectrum between the source and target domains. Third, the Fourier transform is computation-efficient and accelerates the image processing speed, as the multiplication in the frequency domain is equivalent to the convolution operation in the time domain [40].

In addition, the frequency spectrum of deep convolutional features also achieves significant improvements on various tasks. FcaNet proposes multispectral channel attention by generalizing the compression of the channel attention mechanism in the frequency domain, which achieves performance improvements in image classification, object detection, and image segmentation. FACMA [42] introduces a frequency-aware cross-modality attention module for feature learning and increases RGB-D saliency detection results. Deep frequency filtering (DFF) [20] learns the domain-generalizable features by modulating the frequency components of different transfer difficulties across domains to improve the generalization ability of deep models.

This article focuses on the frequency domain analysis based on deep convolutional features for remote sensing image registration. We propose an adaptive F3Net to identify useful frequency components, enhancing feature invariance and discriminability and improving the performance of cross-modal remote sensing image registration.

III. Method

A. Overview

This article proposes an adaptive F3Net for cross-modal remote sensing image matching. As shown in Fig. 2, F3Net extracts multilevel features from optical and SAR images, respectively. Meanwhile, F3Net designs the frequency feature filtering (F3) module for frequency filtering based on multilevel features. To further enhance the matching performances, we adopt an FA module to select important frequency components in the frequency feature filtering process. Finally, F3Net finds the matching location through similarity computation based on the filtered deep features. The details of network structure and optimization are given as follows.

B. Network Structure

1) Network Backbone: F3Net designs a feature extraction module based on several convolutional blocks for deep feature learning. In the shallow layers, the convolutional block consists of convolutional layers (Conv), the instance normalization

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layer (IN), and the activation function of rectified linear unit (ReLU). In the feature learning process, the size of feature maps is consistent with the size of the input image to maintain high spatial resolution and matching accuracy.

2) Frequency Feature Filtering Module: We insert the F3 module in deep layers to enhance the feature learning. Specifically, the extracted features in the previous layer are divided into two parts along the channel dimension and then input for spatial and frequency feature learning, respectively. As shown in Fig. 2, the F3 module learning process can be formulated as follows:

$$x = [x_s, x_f] \tag{1}$$

where $x_s \in R^{H\times W\times 0.5C}$ represents the input of the spatial feature learning and $x_f \in R^{H\times W\times 0.5C}$ represents the input of the frequency feature learning.

The output of the spatial and frequency feature learning branches can be represented as follows:

$$\begin{aligned} y_s &= F_s(x_s) + \mathrm{Conv}(x_f) \\ y_f &= F_f(x_f) + \mathrm{Conv}(x_s) \end{aligned} \tag{2}$$

where F_s and F_f are spatial and frequency feature learning, respectively, Conv represents convolutional operator, and y_s and y_f are their corresponding outputs.

The fused spatial-frequency features can be obtained as follows:

$$y = y_s \oplus_c y_f \tag{3}$$

where y is the fused spatial-frequency feature and $\oplus_c$ represents the channel concatenation.

It should be noted that we not only perform feature fusion based on the output of the spatial and frequency feature learning branches [as shown in (3)] but also conduct the feature interaction between these two branches during the spatial and frequency feature learning [as shown in (2)], which can further promote the spatial-frequency features fusion and enhance the robustness and discrimination of features.

The F3 module utilizes the convolutional layer to capture the spatial features in the spatial domain and learn frequency features through multiscale frequency filtering in the frequency domain. In the frequency feature learning, the input feature is first transformed into the frequency domain through the FFT operator. After that, this article conducts the frequency filtering through convolution operators based on the transformed frequency features. In addition, this article designs an FA learning module to adaptively select the critical frequency information. In this way, the F3 module can adaptively filter out some frequency and noise information detrimental to generalization while retaining the crucial frequencies shared in the heterogeneous images to improve the matching performances. Then, the filtered frequency features are transformed into the spatial domain through the inverse fast Fourier transformation (IFFT). We will present the details of the frequency feature learning branch.

3) Multiscale Frequency Filtering: To reduce the computation cost, we apply a convolutional layer to decrease the channel of the input features before the frequency feature learning

$$x'_f = \mathrm{Conv}(x_f) \tag{4}$$

where Conv is the convolutional operator.

This article designs a multiscale frequency feature filtering, as shown in Fig. 2 (bottom). The frequency feature learning branch contains a GF3 branch and an LF3 branch. GF3 performs frequency filtering on the global spectrum features, and LF3 conducts frequency filtering on the stacked spectrum of local image regions

$$\begin{aligned} x^{\mathrm{GF3}} &= F_{\mathrm{GF3}}\left(x'_f\right) \\ x^{\mathrm{LF3}} &= F_{\mathrm{LF3}}\left(x'_f\right) \end{aligned} \tag{5}$$

where F_{GF3} and F_{LF3} represent the global and local frequency feature filtering, respectively, and x^{GF3} and x^{LF3} are outputs of the GF3 and LF3 branches, respectively.

We will present the details of GF3 as an example. Specifically, we transform feature x'_f into the frequency domain through FFT and perform the convolution operator on the frequency features

$$\begin{aligned} X_F &= \mathrm{FFT}\left(x'_f\right) \\ X'_F &= \mathrm{Conv}(X_F) \end{aligned} \tag{6}$$

where FFT represents the FFT operator. Specifically, the details of the FFT can be formulated as follows:

$$X_F(x, y) = \sum_{h=0}^{H-1}\sum_{w=0}^{W-1} x'_f(h, w) e^{-j2\pi\left(\frac{xh}{H}+\frac{yw}{W}\right)} \tag{7}$$

where H and W are width and height of the input feature map x'_f, respectively.

4) FA Learning: We adopt frequency attention learning to learn the important frequency information adaptively

$$X_{\mathrm{AF}} = \mathrm{FA}\left(X'_F\right) \tag{8}$$

where FA is the frequency attention learning operator and X_{AF} is the output of the FA learning module.

This article combines the average pooling AP and max pooling MP in the FA module. It can be formulated as follows:

$$\begin{aligned} X_F^{\mathrm{map}} &= \mathrm{AP}\left(X'_F\right) \oplus_c \mathrm{MP}\left(X'_F\right) \\ \mathrm{AW}_F &= \mathrm{Conv}\left(X_F^{\mathrm{map}}\right) \\ X_{\mathrm{AF}} &= X'_F \otimes \mathrm{AW}_F \end{aligned} \tag{9}$$

where AW_F is the attention weight of frequency features and $\otimes$ represents the elementwise multiply.

Then, the learned frequency feature is transformed into the spatial domain through the IFFT

$$x^{\mathrm{GF3}} = \mathrm{IFFT}(X_{\mathrm{AF}}) \tag{10}$$

where IFFT represents the IFFT operator. Specifically, the details of the IFFT can be formulated as follows:

$$x^{\mathrm{GF3}}(h, w) = \frac{1}{HW}\sum_{h=0}^{H-1}\sum_{w=0}^{W-1} X_{\mathrm{AF}}(x, y) e^{j2\pi\left(\frac{xh}{H}+\frac{yw}{W}\right)}. \tag{11}$$

The LF3 is similar to the GF3. LF3 first splits the input features into four parts and acquires the stacked local features

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through channel concatenation. Then, LF3 conducts frequency feature filtering on the local features and stitches the filtered local features into the whole feature map.

5) Frequency Feature Fusion: Finally, this article fuses the input feature, the filtered global frequency feature, and the filtered local frequency feature for more accurate remote sensing image matching

$$y_f = F_{\text{fusion}}\left(x'_f, x^{\text{GF3}}, x^{\text{LF3}}\right) \tag{12}$$

where F_{fusion} means the fusion operator, which first sums these features and then performs the convolutional operator.

C. Network Optimization

This article adopts a template-matching approach for remote sensing image registration based on the extracted features. Suppose that the optical and SAR images are I_{opt} and I_{sar}, their corresponding image sizes are $h_1 \times w_1$ and $h_2 \times w_2$, and the extracted features are f_{opt} and f_{sar}. The template matching approach computes the similarity of the features of optical and SAR images and finds the matching position with the maximum similarity.

The similarity of features is calculated as follows:

$$S_{i,j} = \sum_{m=i,\ldots,T_h} \sum_{l=j,\ldots,T_w} \frac{f_{\text{opt}}(m,l)\, f_{\text{sar}}(m,l)}{\|f_{\text{opt}}\|_2 \|f_{\text{sar}}\|_2} \tag{13}$$

where $S_{i,j}$ represents the similarity of optical and SAR images in the location (i, j), and $T_h = i + h_2 - 1$ and $T_w = j + w_2 - 1$ represent the upper bounds of the template window applied to the reference image.

This article adopts matching loss, L_m, fine similarity loss, L_{fine}, and similarity peak loss, L_{peak}, for network optimization. The matching loss aims to constrain the similarity near the ground truth, gt, which is larger than other locations. The fine similarity loss expects the Gaussian distribution of the ground truth and corresponding similarity to be consistent. The similarity peak loss aims to constrain the maximum similarity in the matching location, which is more significant than the mean similarity in all locations.

1) Matching Loss: The matching loss expects the positive positions to have a significant similarity and the negative positions to have a slight similarity. It can be represented as follows:

$$L_m = (S_{\text{neg}} + 1)^2 + (1 - S_{\text{pos}})^2 \tag{14}$$

where S_{pos} is the similarity of the positive positions near the ground truth, such as 5×5 neighborhood of the ground truth, and S_{neg} is the similarity of the negative positions. To accelerate the network convergence, this article introduces the hard negative mining strategy, where the negative samples with the top 25 largest image similarity are used for network optimization.

2) Fine Similarity Loss: The matching loss merely focuses on separating the positive positions from the negatives. To further enhance the network optimization and find the fine-matching location, this article utilizes the fine similarity loss to constrain the similarity of the positive region. The similarity at the matching position should be the maximum, and the similarities at its neighboring positions should be inversely proportional to their distances from the gt. It means that the closer the ground truth, the greater the feature similarity. The fine similarity loss, L_{fine}, is expressed as follows:

$$\begin{aligned} G_{\text{gt}} &= \text{Gaussian(gt)} \\ L_{\text{fine}} &= \left(\text{top}k(G_{\text{gt}}) - S_{\text{pos}}^{\text{top}k}\right)^2 \end{aligned} \tag{15}$$

where G_{gt} is the soft labels obtained by Gaussian blurring around the ground truth alignment position, $\text{top}k(G_{\text{gt}})$ means select the top k value of G_{gt}, $S_{\text{pos}}^{\text{top}k}$ represents the feature similarity in the corresponding positions of $\text{top}k(G_{\text{gt}})$, and $k = 9$ in this article.

3) Similarity Peak Loss: In addition, to avoid the feature similarity in different positions tends to be consistent and results in the failure to find the matching position, this article uses the similarity peak loss, L_{peak}, for network optimization. It expects the maximize similarity to be far larger than the mean similarity, which can be formulated as follows:

$$L_{\text{peak}} = 2 - (\max(S) - \text{mean}(S)) \tag{16}$$

where $\max(S)$ and $\text{mean}(S)$ are the maximize and the mean of the similarity map, respectively.

The final optimization loss for network training can be represented as follows:

$$L = L_m + \alpha L_{\text{fine}} + \beta L_{\text{peak}} \tag{17}$$

where α and β are the weighting factors of the fine similarity loss and similarity peak loss, respectively.

IV. Experiments

This part verifies the effectiveness of our proposed F3Net on many multimodal remote sensing images. We will present the dataset introduction, implementation details, experimental results, and analysis.

A. Datasets

The experiments are conducted on the SEN1-2 dataset. The SEN1-2 dataset contains many optical and SAR image pairs generated from Sentinel-1 (SEN-1) and Sentinel-2 (SEN-2). SEN-1 was developed by the European Space Agency (ESA) and consists of two satellites: Sentinel-1A and Sentinel-1B. These satellites carry C-band SAR sensors, which can provide dual-polarization SAR images under all-weather, day, and night conditions. SEN-2 also includes two satellites (Sentinel-2A and Sentinel-2B) developed by ESA, which can provide multispectral remote sensing images. The SEN1-2 dataset has four subsets corresponding to different seasons: spring, summer, autumn, and winter. In the experiments, 11 106 image pairs were used for training, and 1587 image pairs were used for testing. As shown in Fig. 3, each image size is 256×256, and the image resolution is 10 m. The SAR images are cropped to 128×128. The cropped SAR images are then used as templates for multimodal image matching with the corresponding reference images (optical images), and their corresponding top-left point position can be viewed as the ground truth of matching. The coordinate value of the

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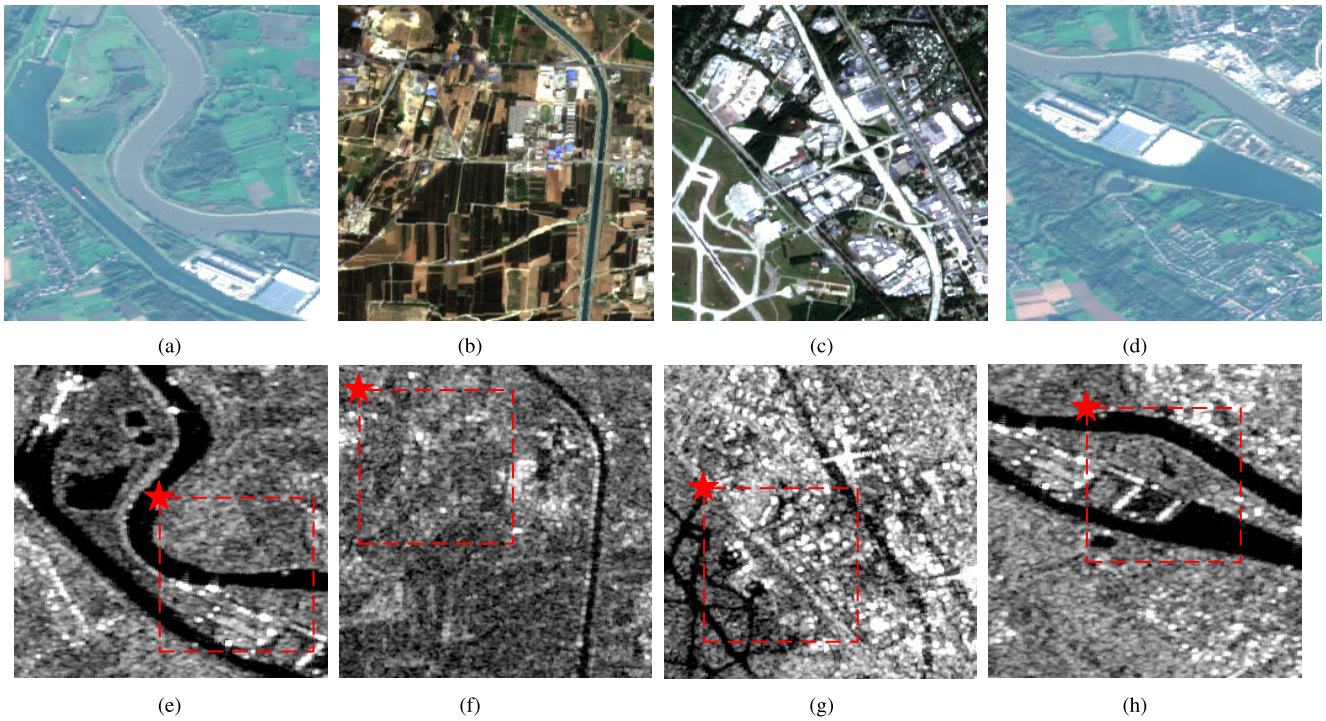


Fig. 3. Optical and SAR images in the SEN1-2 dataset. The red box represents a cropped SAR template for image matching. The star indicates the matching location of the ground truth. (a)–(d) Optical image. (e)–(h) SAR image.

matching position is from 0 to 127 in horizontal and vertical directions. The main challenges of the image matching and registration on the SEN1-2 dataset are presented as follows.

1) *Significant Appearance Differences:* There are significant appearance differences between optical and SAR images due to different imaging mechanisms. It is challenging to learn robust features that are invariant to image modality changes for finding the matching samples and maintain high discrimination for separating nonmatching samples simultaneously.
2) *Speckle Noise Interference:* SAR images contain a lot of Speckle noise, which impedes feature learning and representation of SAR images.
3) *Large Offsets:* There are large offsets between multimodal images, which will greatly increase the difficulty of image matching and registration. The large offsets mean that the deep model should identify the matching location through a wide range of searches and distinguish the confusion and interference from many nonmatching locations.

We also tested the matching performance of the trained deep model on the other 12 pairs of multimodal remote sensing images to verify the generalization of the proposed method.

B. Implement Details

The experiments were conducted using an Nvidia RTX3090 GPU and implemented using the PyTorch framework. The Adam optimizer was used to train the network. The training epochs is 30, with a learning rate of 0.001 and a batch size of 2.

C. Evaluation Metrics

This article adopts the root mean squared error (RMSE) and matching accuracy as evaluation metrics. RMSE measures the root mean squared error between the predicted matching locations and the ground truth between the template image (SAR image) and the reference image (optical image). The RMSE is calculated as

$$\mathrm{RMSE} = \sqrt{\frac{1}{N}\sum_{l=1}^{N}\left[\left(i_l - i_l'\right)^2 + \left(j_l - j_l'\right)^2\right]} \tag{18}$$

where (i_l', j_l') represents the matching coordinates predicted by the network for the l_{th} image pairs, (i_l, j_l) denotes the matching positions of the ground truth, and N is the total number of test images.

The matching accuracy is the ratio of successful matching images where the RMSE is below a threshold. For example, Acc_θ means the matching accuracy when the RMSE $\leq \theta$.

D. Comparison Experiments

This part compares the matching results of different methods, such as the traditional template matching method (MI [22]), handcrafted local feature matching methods (SIFT [5], Harris [23], and CFOG [27]), deep local feature learning method (HardNet [28]), and deep global feature learning methods (deep dense feature network (DDFN) [43], MCGF [32], and GFTM [33]). The experimental results are shown in Table I.

From these experimental results, we can draw the following conclusions. First, the proposed F3Net achieves the lowest error and the highest matching accuracy based on different

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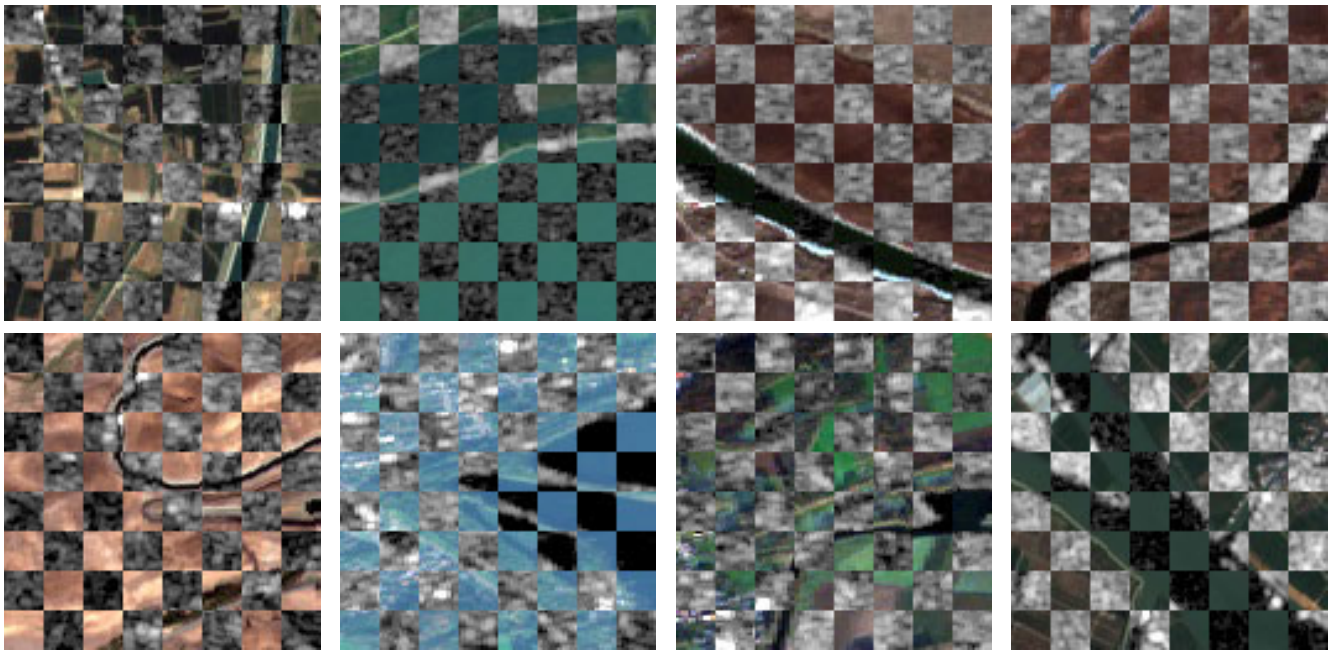


Fig. 4. Checkboard images in the matching and overlapping regions of optical and SAR images.

TABLE I
COMPARISON EXPERIMENTS ON THE SEN1-2 DATASET

Method	RMSE	Acc_1	Acc_2	Acc_3
MI [22]	17.75	33.0 %	48.6%	58.3%
SIFT [5]	70.39	0%	0.1%	0.3%
Harris [23]	20.12	15.5%	21.5%	27.1%
CFOG [27]	13.29	38.1 %	54.9%	61.1%
HardNet [28]	15.46	31.6%	52.4%	65.0%
DDFN [43]	4.62	47.2 %	62.5%	69.7%
MCGF [32]	4.36	49.8 %	64.7%	72.8%
GFTM [33]	3.69	49.5%	71.1%	81.1%
F3Net (Our)	**2.33**	**54.3%**	**95.8%**	**97.3%**

error thresholds. Second, the deep local and global feature learning methods acquire better matching performances than the traditional methods. For example, the best RMSE result of the low-level handcrafted features is 13.29, the optimal RMSE result of other deep learning methods is 3.69, and the RMSE result of our proposed method is 2.33. In addition, the best Acc_3 result of the low-level handcrafted features is 61.1%, the optimal Acc_3 result of other deep learning methods is 81.1%, and the Acc_3 result of our proposed method is 97.3%. The primary reason is that deep learning methods have stronger representation abilities than image intensity and handcrafted features. Third, global feature learning methods perform better results than local feature methods. Because of the limited perception of image content within a local image patch, deep local feature learning methods are vulnerable to the interference and confusion of similar local patterns, leading to a large number of false matching point pairs. The rich contextual information in the global features can mitigate the confusion caused by local image similarity and boost the matching accuracy. For example, the Acc_3 result of the deep local feature learning method HardNet is 65.0%, while the Acc_3 result of the deep global feature learning method is 69.7% in DDFN, 72.8% in MCGF, 81.1% in GFTM, and 97.3% in our F3Net. Finally, the proposed method presents significant advantages over other deep global feature learning methods. The main reason is that our method not only captures pixelwise dense deep global features but also conducts frequency feature filtering. On the one hand, frequency feature filtering can adaptively select the important frequency information shared to identify the matching multimodal location and discriminative for distinguishing the nonmatching location. On the other hand, the convolutional operator will encounter limitations of the local respective field in the space domain. However, the frequency feature filtering can take advantage of nonlocal respective fields by frequency modulation for feature learning. Specifically, F3Net achieves significant performance improvements compared with existing methods. Specifically, the smallest accuracy gains of Acc_3, Acc_2, and Acc_1 are 16.2%, 24.7%, and 4.8%, respectively. The smallest performance improvement of RMSE is 1.36 (the registration error is decreased by 36.86%).

E. Visualization

This part displays several checkboard images in the matching and overlap regions of optical and SAR images. As shown in Fig. 4, the intersections are very smooth between optical and SAR images, which also demonstrates the significant advantages of the proposed method on multimodal image registration. However, the images with poor texture features will result in a large matching error, as shown in Fig. 5. In the future, we will focus on improving the image matching performances for images with poor textures.

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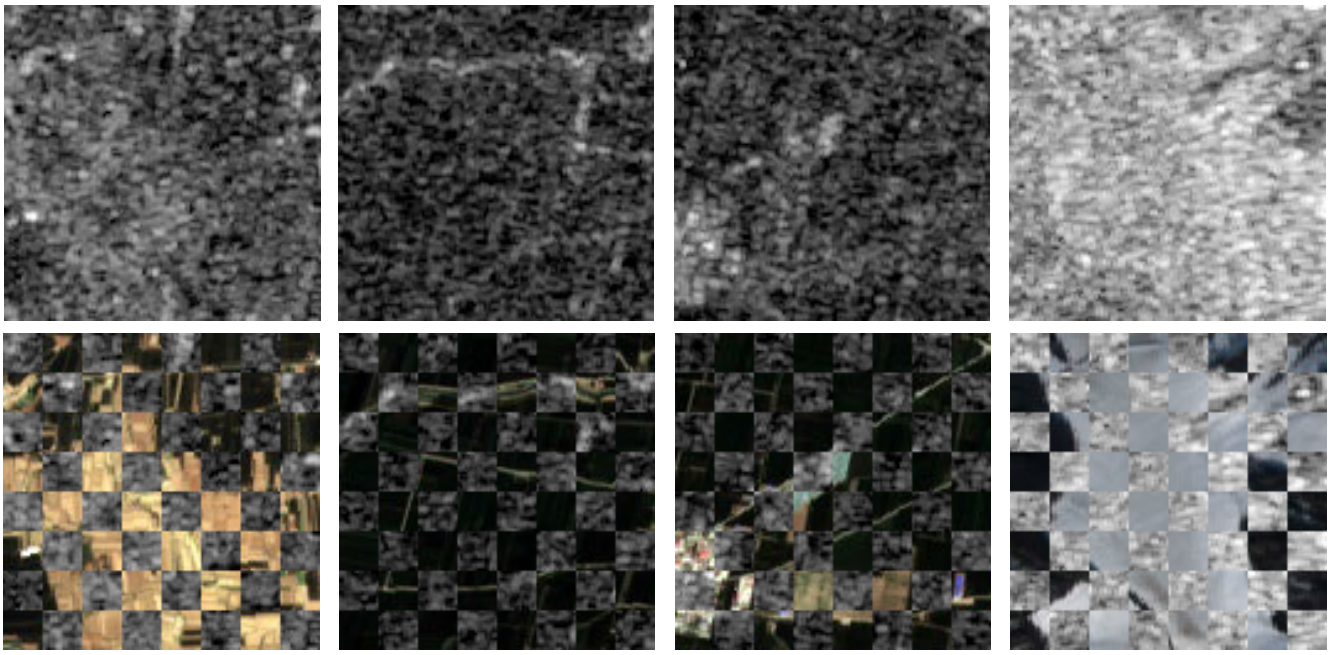


Fig. 5. SAR images (top) and their corresponding checkboard images (bottom). These SAR images have poor texture features, resulting in a large matching error.

We also give the visualization of the extracted features by the general deep convolutional network and our F3Net, as shown in Fig. 6. The general deep convolutional network has the same convolutional structure as F3Net but without the F3 module. It can be observed that the features learned by the general deep convolutional network have fewer distinct details. The reason may be that the convolution operation will smooth the image during feature extraction, causing some fine-grained texture information to be lost. The features extracted by F3Net exhibit more apparent textures and prominent details. These results indicate that F3Net can extract more distinctive features, improving image matching performances.

F. Model Generalization

The deep model should have good generalization performances on other remote sensing image matching. To test the generalization of the proposed F3Net, we directly apply the trained deep network based on the SEN1-2 dataset for other multimodal images. As shown in Fig. 7, there are 12 multimodal image pairs and their corresponding checkboard images of the matching and overlapping regions between the reference and template images. It can be seen that the F3Net acquires better matching performances on these images although there are significant differences between these test multimodal images and training images in the SEN1-2 dataset. These experimental results also show that the proposed F3Net has strong generalization on other multimodal remote sensing images.

G. Ablation Study

1) Effectiveness of F3 Module: This part explores the network structure of the proposed F3Net, and the experimental results are shown in Table II. The "Base" represents the network with a standard deep convolutional model without frequency feature filtering. "F3Net-5" denotes that the fifth layer of the convolutional network inserts the proposed F3 module, "F3Net-5, 6" denotes that the fifth and sixth convolutional layers insert the proposed F3 module, and other experimental settings can be done in a similar manner.

TABLE II
ABLATION EXPERIMENTS ON THE NUMBER OF THE F3 MODULE

Method	$RMSE$	Acc_3
Base	2.82	96.7%
F3Net-5	2.64	96.4%
F3Net-5,6	2.60	97.1%
F3Net-5,6,7	2.58	97.1%
F3Net-5,6,7,8	**2.33**	**97.3%**

The experimental results indicate that as more layers are inserted with the proposed F3 module, the deep network achieves lower RMSE and higher matching accuracy. Specifically, compared with the "Base" model, the matching accuracy is improved by 0.9%, and RMSE is decreased by 0.492 by "F3Net-5,6,7,8." The subsequent experiments will adopt the network structure as the "F3Net-5,6,7,8." These results demonstrate the effectiveness of the proposed frequency feature filter module. The deep features filtered by the F3 module will enhance the robustness of features, leading to performance improvements.

2) Effectiveness of FA Module: This part verifies the effectiveness of the FA module. As shown in Table III, "W/O FA" represents without FA module in the frequency feature filtering. "W FA-Channel" and "W FA-Spatial" represent the network that adopts the channel attention and spatial attention module in the frequency feature filtering, respectively. "W FA-

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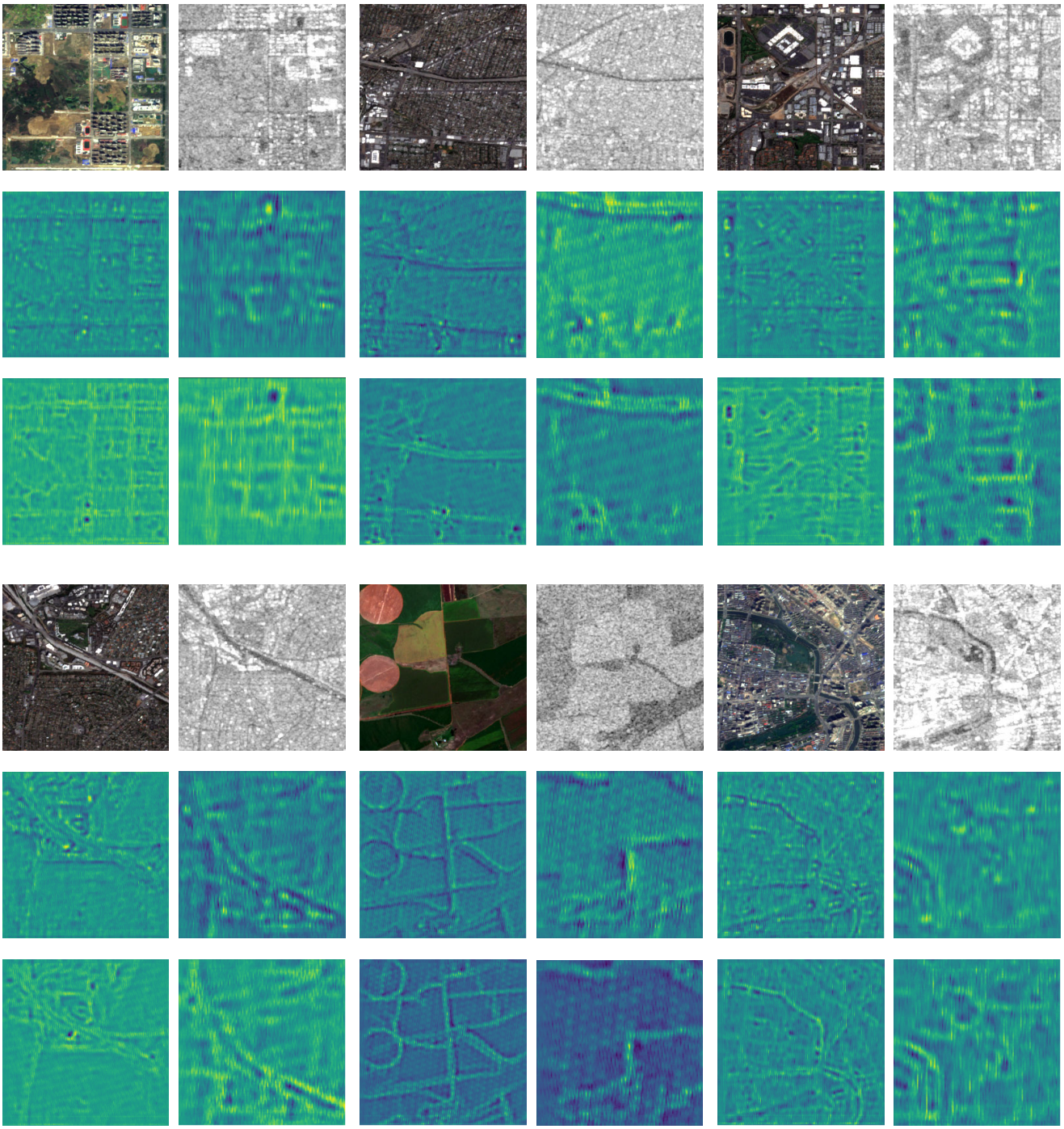


Fig. 6. Multimodal images (in the top row) and their corresponding feature visualizations through the general deep convolutional model (in the middle row) and our F3Net (in the bottom row) in the SEN1-2 dataset.

CBAM" means that the network combines channel and spatial attention in frequency feature filtering.

The experimental results show that the introduced FA module will improve the matching performances. Compared with the results of W/O FA, the matching accuracy is improved by 1.2%, 1.8%, and 0.6%, and RMSE decreases by 0.442, 0.766, and 0.351 through the "W FA-Channel," "W FA-Spatial," and "W FA-CBAM," respectively. The proposed FA module will encourage the feature learning network to focus on more important frequency information for image matching. In addition, spatial attention learning will bring more significant improvements than channel attention. We think the main reason may be that frequency information is mainly distributed spatially in the frequency domain. Spatial attention is more helpful for selecting important frequency information. Thus, we adopt spatial attention learning in the FA module.

3) Effectiveness of the Loss Function: This article utilizes three loss functions for network optimization, such as matching loss, L_m, fine similarity loss, L_{fine}, and similarity

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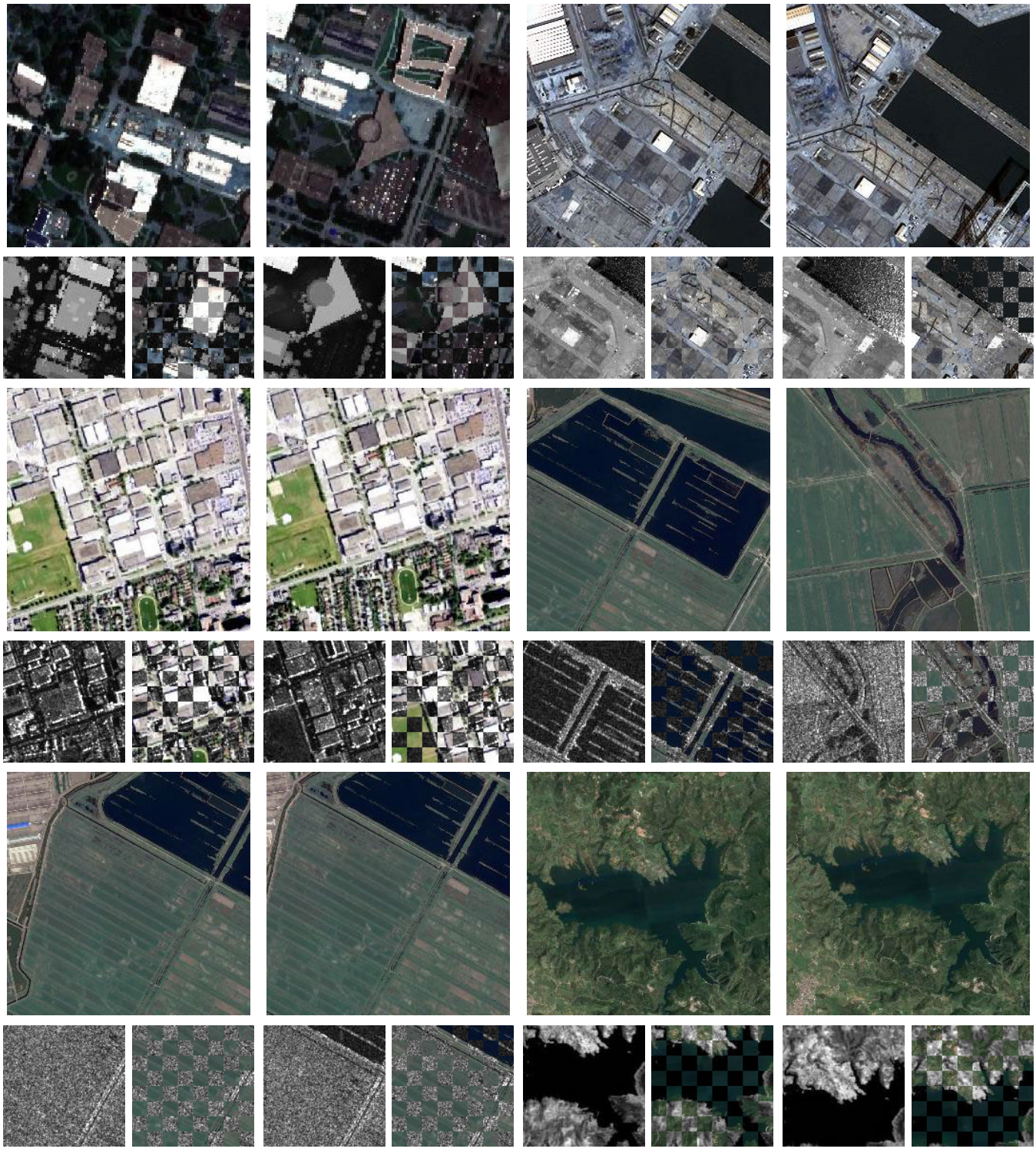


Fig. 7. Twelve pairs of multimodal images (reference image on the top and template images on the left bottom) and the corresponding checkboard images (on the right bottom) in the matching and overlapping regions of the reference and template images.

peak loss, L_{peak}. This part tests the effectiveness of different loss functions.

The matching loss is a common basic optimization loss for image matching and registration, which can pull close the features of the matching locations and push away the features of nonmatching locations. It is effective in acquiring good results for most multimodal images. As shown in Table IV, the Acc_3 result of L_m is 92.8%. However, the locations close to the ground truth are viewed as matching locations to stabilize the training process. The deep network that is merely trained on the matching loss is challenging to find a more precise matching position. To achieve more accurate results, the fine similarity loss constrains the closer the ground truth, the greater the feature similarity. In addition, to avoid the feature

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TABLE III
ABLATION EXPERIMENTS ON THE FA MODULE

Module	$RMSE$	Acc_3
W/O FA	3.09	95.5%
W FA-Channel	2.65	96.7%
W FA-Spatial	**2.33**	**97.3%**
W FA-CBAM	2.74	96.1%

TABLE IV
EXPERIMENTAL RESULTS BASED ON DIFFERENT LOSS FUNCTIONS

L_m	L_{peak}	L_{fine}	$RMSE$	Acc_3
✓			2.97	92.8%
✓	✓		2.65	96.2%
✓		✓	2.69	97.0%
✓	✓	✓	**2.33**	**97.3%**

TABLE V
EXPERIMENTAL RESULTS BASED ON DIFFERENT HYPERPARAMETERS OF α AND β

α	β	$RMSE$	Acc_3
0.5	0.5	3.53	96.5%
1	0.5	4.27	94.0%
0.5	1	2.73	96.4%
1	1	**2.33**	**97.3%**

similarity in different positions being consistent and resulting in the failure to find the matching position, this article adopts the similarity peak loss for network optimization. It expects the maximize similarity to be far larger than the mean similarity. Experimental results show that the fine similarity loss and the similarity peak loss can enhance network optimization and further improve matching performances. Specifically, the matching accuracy is improved by 3.4% and 4.2% by L_{peak} and L_{fine}, respectively. When we combine L_{peak} and L_{fine} for network training, the matching accuracy is improved by 4.5%, and the RMSE is decreased by 0.64. These experimental results have shown the effectiveness of the optimization losses. We also test the results based on different hyperparameters of α and β, as shown in Table V. α and β are weight coefficients of L_{fine} and L_{peak}, respectively. It can be seen that F3Net achieves the best performances when $\alpha = 1$ and $\beta = 1$. Thus, we adopt the optimal hyperparameters in experiments.

V. CONCLUSION

This article proposes an adaptive F3Net for multimodal remote sensing image registration. F3Net performs frequency filtering on multilevel deep features, which can explicitly explore the useful frequency information and take advantage of the nonlocal receptive fields by frequency modulation for more robust feature learning and boosting image registration performances. F3Net also designs an FA module to adaptive select the essential frequency components between multimodal images. In addition, F3Net adopts multiscale frequency filtering fusion, which combines GF3 and LF3 based on the image-level spectrum and the stacked local image regions, respectively. Experimental results indicate that our F3Net can achieve advantages over traditional methods, deep local matching methods, and other deep global feature learning methods on multimodal image registration. However, the introduced F3 module and the FA module in F3Net will increase the training and computation costs. In future work, we will focus on improving the training efficiency of the deep F3Net and explore more effective frequency feature learning methods for representation augmentation on images with poor textures.

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